

Tests a effectuer pour vérifier le bon fonctionnement du VPN avec les client

1) Le VPN est connecter, regarder les logs sur serveurs pour voir ce que les logs racontent.

Consultation des journaux (logs) : `tail -f /var/log/openvpn.log`

Effectuer commande pour chaque client :

- Windows 

```
root@SRV-OpenVPN:~# tail -f /var/log/openvpn.log
37.166.73.125:3774 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key:
253 bits X25519
37.166.73.125:3774 [client1 Windows] Peer Connection Initiated with [AF_INET]37.166.73.125:3774
client1 Windows/37.166.73.125:3774 MULTI_sva: pool returned IPv4=10.8.0.6, IPv6=(Not enabled)
client1 Windows/37.166.73.125:3774 MULTI: Learn: 10.8.0.6 -> client1 Windows/37.166.73.125:3774
client1 Windows/37.166.73.125:3774 MULTI: primary virtual IP for client1 Windows/37.166.73.125:3774: 10.8.0.6
client1 Windows/37.166.73.125:3774 SENT CONTROL [client1 Windows]: 'PUSH_REPLY,route 172.22.0.0 255.255.0.0,route 10.8.0.1,topology net30,ping 10,ping-restart 1
20,ifconfig 10.8.0.6 10.8.0.5,peer-id 1,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500' (status=1)
client1 Windows/37.166.73.125:3774 PUSH: Received control message: 'PUSH_REQUEST'
client1 Windows/37.166.73.125:3774 Data Channel: cipher 'AES-256-GCM', peer-id: 0
client1 Windows/37.166.73.125:3774 Timers: ping 10, ping-restart 240
client1 Windows/37.166.73.125:3774 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

- Linux 

```
client2 Linux/37.166.73.125:3748 CC-EEN exit message received by peer
client2 Linux/37.166.73.125:3748 Delayed exit in 5 seconds
client2 Linux/37.166.73.125:3748 SIGTERM[soft,delayed-exit] received, client-instance exiting
37.166.73.125:3826 VERIFY OK: depth=1, CN=src_vpn
37.166.73.125:3826 VERIFY OK: depth=0, CN=client2 Linux
37.166.73.125:3826 peer info: IV_VER=2.6.14
37.166.73.125:3826 peer info: IV_PLAT=linux
37.166.73.125:3826 peer info: IV_TPNL=1
37.166.73.125:3826 peer info: IV_MTU=1600
37.166.73.125:3826 peer info: IV_NCP=2
37.166.73.125:3826 peer info: IV_CIPHERS=AES-256-GCM:AES-128-GCM:CHACHA20-POLY1305
37.166.73.125:3826 peer info: IV_PROTO=990
37.166.73.125:3826 peer info: IV_L2O_STUB=1
37.166.73.125:3826 peer info: IV_COMP_STUB=1
37.166.73.125:3826 peer info: IV_COMP_STUBv2=1
37.166.73.125:3826 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
37.166.73.125:3826 TLS: tls_multi_process: initial untrusted session promoted to trusted
37.166.73.125:3826 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key:
253 bits X25519
37.166.73.125:3826 [client2 Linux] Peer Connection Initiated with [AF_INET]37.166.73.125:3826
client2 Linux/37.166.73.125:3826 MULTI_sva: pool returned IPv4=10.8.0.14, IPv6=(Not enabled)
client2 Linux/37.166.73.125:3826 MULTI: Learn: 10.8.0.14 -> client2 Linux/37.166.73.125:3826
client2 Linux/37.166.73.125:3826 MULTI: primary virtual IP for client2 Linux/37.166.73.125:3826: 10.8.0.14
client2 Linux/37.166.73.125:3826 SENT CONTROL [client2 Linux]: 'PUSH_REPLY,route 172.22.0.0 255.255.0.0,route 10.8.0.1,topology net30,ping 10,ping-restart 120,i
fconfig 10.8.0.14 10.8.0.13,peer-id 0,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500' (status=1)
client2 Linux/37.166.73.125:3826 Data Channel: cipher 'AES-256-GCM', peer-id: 0
client2 Linux/37.166.73.125:3826 Timers: ping 10, ping-restart 240
client2 Linux/37.166.73.125:3826 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

- IOS 

```
37.166.73.125:3828 VERIFY OK: depth=1, CN=src_vpn
37.166.73.125:3828 VERIFY OK: depth=0, CN=client3 IOS
37.166.73.125:3828 peer info: IV_VER=3.11.1
37.166.73.125:3828 peer info: IV_PLAT=ios
37.166.73.125:3828 peer info: IV_NCP=2
37.166.73.125:3828 peer info: IV_TPNL=1
37.166.73.125:3828 peer info: IV_PROTO=8094
37.166.73.125:3828 peer info: IV_MTU=1600
37.166.73.125:3828 peer info: IV_CIPHERS=AES-128-CBC:AES-192-CBC:AES-256-CBC:AES-128-GCM:AES-192-GCM:AES-256-GCM:CHACHA20-POLY1305
37.166.73.125:3828 peer info: IV_AUTO_SESS=1
37.166.73.125:3828 peer info: IV_GUI_VER=net.openvpn.connect.ios.3.7.3-7004
37.166.73.125:3828 peer info: IV_SS0=webauth,crtxt
37.166.73.125:3828 peer info: IV_BS64DL=1
37.166.73.125:3828 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
37.166.73.125:3828 TLS: tls_multi_process: initial untrusted session promoted to trusted
37.166.73.125:3828 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key:
253 bits X25519
37.166.73.125:3828 [client3 IOS] Peer Connection Initiated with [AF_INET]37.166.73.125:3828
client3 IOS/37.166.73.125:3828 MULTI_sva: pool returned IPv4=10.8.0.18, IPv6=(Not enabled)
client3 IOS/37.166.73.125:3828 MULTI: Learn: 10.8.0.18 -> client3 IOS/37.166.73.125:3828
client3 IOS/37.166.73.125:3828 MULTI: primary virtual IP for client3 IOS/37.166.73.125:3828: 10.8.0.18
client3 IOS/37.166.73.125:3828 SENT CONTROL [client3 IOS]: 'PUSH_REPLY,route 172.22.0.0 255.255.0.0,route 10.8.0.1,topology net30,ping 10,ping-restart 120,ifcon
fig 10.8.0.18 10.8.0.17,peer-id 2,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500' (status=1)
client3 IOS/37.166.73.125:3828 PUSH: Received control message: 'PUSH_REQUEST'
client3 IOS/37.166.73.125:3828 Data Channel: cipher 'AES-256-GCM', peer-id: 0
client3 IOS/37.166.73.125:3828 Timers: ping 10, ping-restart 240
client3 IOS/37.166.73.125:3828 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

- Android 

```

37.166.73.125:3830 VERIFY OK: depth=1, CN=srv_vpn
37.166.73.125:3830 VERIFY OK: depth=0, CN=client4 Android
37.166.73.125:3830 peer info: IV_VER=3.11.1
37.166.73.125:3830 peer info: IV_PLAT=android
37.166.73.125:3830 peer info: IV_NCP=2
37.166.73.125:3830 peer info: IV_TCPNL=1
37.166.73.125:3830 peer info: IV_PROTO=8094
37.166.73.125:3830 peer info: IV_MTU=1600
37.166.73.125:3830 peer info: IV_CIPHERS=AES-128-CBC:AES-192-CBC:AES-256-CBC:AES-128-GCM:AES-192-GCM:AES-256-GCM:CHACHA20-POLY1305
37.166.73.125:3830 peer info: IV_AUTO_SESS=1
37.166.73.125:3830 peer info: IV_GUI_VER=net.openvpn.connect.android.3.7.1-10568
37.166.73.125:3830 peer info: IV_SSO=webauth,ctxtext
37.166.73.125:3830 peer info: IV_BSG4DL=1
37.166.73.125:3830 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
37.166.73.125:3830 TLS: tls_multi_process: initial untrusted session promoted to trusted
37.166.73.125:3830 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key:
259 bits X25519
37.166.73.125:3830 [client4 Android] Peer Connection Initiated with [AF_INET]37.166.73.125:3830
client4 Android/37.166.73.125:3830 MULTI_svr: pool returned IPv4=10.8.0.10, IPv6=(Not enabled)
client4 Android/37.166.73.125:3830 MULTI: Learn: 10.8.0.10 -> client4 Android/37.166.73.125:3830
client4 Android/37.166.73.125:3830 MULTI: primary virtual IP for client4 Android/37.166.73.125:3830: 10.8.0.10
client4 Android/37.166.73.125:3830 SENT CONTROL [client4 Android]: 'PUSH_REPLY,route 172.22.0.0 255.255.0.0,route 10.8.0.1,topology net30,ping 10,ping-restart 1
20,ifconfig 10.8.0.10 10.8.0.9,peer-id 3,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1500' (status=1)
client4 Android/37.166.73.125:3830 PUSH: Received control message: 'PUSH_REQUEST'
client4 Android/37.166.73.125:3830 Data Channel: cipher 'AES-256-GCM', peer-id: 0
client4 Android/37.166.73.125:3830 Timers: ping 10, ping-restart 240
client4 Android/37.166.73.125:3830 Protocol options: protocol-flags cc-exit tls-ekm dyn-tls-crypt

```

2) Sur PC client : quelle conf ip ? Est-ce qu'il prend bien une adresse IP en 10.8.0.x ? Est-ce que il est possible de ping sont peer (adresse ip du serveur) 10.8.0.1 ?

Windows :

Le client prend bien une IP dans le tunnel VPN : 10.8.0.6

Carte inconnue Connexion au réseau local :

```

Suffixe DNS propre à la connexion. . . . :
Adresse IPv6 de liaison locale. . . . . : fe80::2e18:1728:35a7:8838%12
Adresse IPv4. . . . . : 10.8.0.6
Masque de sous-réseau. . . . . : 255.255.255.252
Passerelle par défaut. . . . . :

```

Le client ping bien le serveur : [ping 10.8.0.1](#)

```

C:\Users\paulg>ping 10.8.0.1

Envoi d'une requête 'Ping' 10.8.0.1 avec 32 octets de données :
Réponse de 10.8.0.1 : octets=32 temps=33 ms TTL=64
Réponse de 10.8.0.1 : octets=32 temps=35 ms TTL=64
Réponse de 10.8.0.1 : octets=32 temps=43 ms TTL=64
Réponse de 10.8.0.1 : octets=32 temps=72 ms TTL=64

Statistiques Ping pour 10.8.0.1:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 33ms, Maximum = 72ms, Moyenne = 45ms

```

Le serveur ping bien le client Windows : [ping 10.8.0.6](#)

```

root@SRV-OpenVPN:~# ping 10.8.0.6
PING 10.8.0.6 (10.8.0.6) 56(84) bytes of data.
64 bytes from 10.8.0.6: icmp_seq=1 ttl=128 time=78.6 ms
64 bytes from 10.8.0.6: icmp_seq=2 ttl=128 time=164 ms
64 bytes from 10.8.0.6: icmp_seq=3 ttl=128 time=27.3 ms
64 bytes from 10.8.0.6: icmp_seq=4 ttl=128 time=78.6 ms
^C
--- 10.8.0.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 27.337/87.029/163.637/48.925 ms

```

Linux :

Le client prend bien une IP dans le tunnel VPN : 10.8.0.14

```

4: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 500
    link/none
    inet 10.8.0.14 peer 10.8.0.13/32 scope global noprefixroute tun0
        valid_lft forever preferred_lft forever
    inet6 fe80::1e0:41d4:e110:87b2/64 scope link stable-privacy
        valid_lft forever preferred_lft forever

```

Le client ping bien le serveur : [ping 10.8.0.1](#)

```

root@debian:~# ping 10.8.0.1
PING 10.8.0.1 (10.8.0.1) 56(84) bytes of data.
64 bytes from 10.8.0.1: icmp_seq=1 ttl=64 time=29.1 ms
64 bytes from 10.8.0.1: icmp_seq=2 ttl=64 time=30.2 ms
64 bytes from 10.8.0.1: icmp_seq=3 ttl=64 time=51.8 ms
64 bytes from 10.8.0.1: icmp_seq=4 ttl=64 time=33.8 ms
^C
--- 10.8.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 29.073/36.234/51.811/9.161 ms

```

Le serveur ping bien le client Windows : [ping 10.8.0.14](#)

```

root@SRV-OpenVPN:~# ping 10.8.0.14
PING 10.8.0.14 (10.8.0.14) 56(84) bytes of data.
64 bytes from 10.8.0.14: icmp_seq=1 ttl=64 time=30.8 ms
64 bytes from 10.8.0.14: icmp_seq=2 ttl=64 time=39.5 ms
64 bytes from 10.8.0.14: icmp_seq=3 ttl=64 time=44.6 ms
64 bytes from 10.8.0.14: icmp_seq=4 ttl=64 time=36.3 ms
^C
--- 10.8.0.14 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 30.833/37.787/44.578/4.990 ms

```

3) Quelles routes sur ton PC vpn ? "route print" => ça permet de comprendre ce qui passe où ; si tu as fais un push route, est-il effectif

Windows :

172.22.0.0	255.255.0.0	10.8.0.5	10.8.0.6	257
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Linux :

```
root@debian:~# ip route
default via 192.168.248.2 dev ens33 proto dhcp src 192.168.248.165 metric 100
10.8.0.1 via 10.8.0.13 dev tun0 proto static metric 50
10.8.0.13 dev tun0 proto kernel scope link src 10.8.0.14 metric 50
89.87.185.143 via 192.168.248.2 dev ens33 proto static metric 50
172.22.0.0/16 via 10.8.0.13 dev tun0 proto static metric 50
192.168.248.0/24 dev ens33 proto kernel scope link src 192.168.248.165 metric 100
192.168.248.2 dev ens33 proto static scope link metric 50
```

4) Est-ce, étant dans la DMZ, tu ping la patte DMZ du firewall => 172.22.30.1

Depuis le serveur [ping 172.22.30.1](#)

```
SRV-OpenVPN:~# nano /etc/openvpn/server/server.conf
SRV-OpenVPN:~# ping 172.22.30.1
PING 172.22.30.1 (172.22.30.1) 56(84) bytes of data.
64 bytes from 172.22.30.1: icmp_seq=1 ttl=64 time=0.241 ms
64 bytes from 172.22.30.1: icmp_seq=2 ttl=64 time=0.210 ms
64 bytes from 172.22.30.1: icmp_seq=3 ttl=64 time=0.248 ms
64 bytes from 172.22.30.1: icmp_seq=4 ttl=64 time=0.267 ms
^C
--- 172.22.30.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3019ms
rtt min/avg/max/mdev = 0.210/0.241/0.267/0.020 ms
```

Depuis le client Windows [ping 172.22.30.1](#)

```
C:\Users\paulg>ping 172.22.30.1

Envoi d'une requête 'Ping' 172.22.30.1 avec 32 octets de données :
Réponse de 172.22.30.1 : octets=32 temps=50 ms TTL=63
Réponse de 172.22.30.1 : octets=32 temps=37 ms TTL=63
Réponse de 172.22.30.1 : octets=32 temps=74 ms TTL=63
Réponse de 172.22.30.1 : octets=32 temps=39 ms TTL=63

Statistiques Ping pour 172.22.30.1:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 37ms, Maximum = 74ms, Moyenne = 50ms
```

Depuis le client Linux [ping 172.22.30.1](#)

```
paul@debian:~$ ping 172.22.30.1
PING 172.22.30.1 (172.22.30.1) 56(84) bytes of data.
64 bytes from 172.22.30.1: icmp_seq=1 ttl=128 time=72.0 ms
64 bytes from 172.22.30.1: icmp_seq=2 ttl=128 time=72.2 ms
64 bytes from 172.22.30.1: icmp_seq=3 ttl=128 time=42.2 ms
^C
--- 172.22.30.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 42.181/62.130/72.207/14.106 ms
```

Pour vérifier si le VPN fonctionne a 100%. Ouvrir sur un explorateur internet les sites internet qui doivent être accessible en interne grâce au VPN.

<https://172.22.100.8/zabbix> Windows OK / Linux OK / IOS OK / Android OK

<http://tournoifive.m2l.local/adminer.php> Windows OK / Linux OK / IOS OK / Android OK

<http://172.22.100.15:9000/> Windows OK / Linux OK / IOS OK / Android OK

https://matchplanner.m2l.local/ppe_1/public/index.php Windows OK / Linux OK / IOS OK / Android OK

<http://matchplanner.m2l.local/adminer.php> Windows OK / Linux OK / IOS OK / Android OK

<http://five.resa.m2l.local/> Windows OK / Linux OK / IOS OK / Android OK

http://five.resa.m2l.local/ppe_1_rayane/projet-ppe-foot5-web/frontend/build Windows OK / Linux OK / IOS OK / Android OK

Pour que l' on puisse accéder au site internet des Slam directement avec le nom et pas l'adresse IP

Ajouter dans le fichier « server.conf » :

- Une ligne pour pousser l'adresse IP de ton serveur DNS interne aux clients.

`push "dhcp-option DNS 172.22.100.4"`

- Une ligne pour pousser l'adresse IP de ton serveur DNS de secours aux clients.

`push "dhcp-option DNS 172.22.100.5".`

- Une ligne pour pousser le suffixe de recherche DNS. Cela permet de taper uniquement "matchplanner" au lieu de "matchplanner.m2l.local".

`push "dhcp-option DOMAIN m2l.local"`

```
# Options réseau
keepalive 10 120      # ping entre serveur/clients pour vérifier connexion VPN
persist-key          # conserve clés meme après restart
persist-tun          # conserve interface TUN meme après restart
# push "route 172.22.120.0 255.255.255.0" # route vers le sous réseau (Foot)
# push "route 172.22.130.0 255.255.255.0" # route vers le sous réseau (Tennis)
push "route 172.22.0.0 255.255.0.0" # route plus large (route directement vers le sur-réseau)
push "dhcp-option DNS 172.22.100.4" # pousse l'adresse IP du serveur DNS interne aux clients.
push "dhcp-option DNS 172.22.100.5" # pousse adresse IP du serveur DNS de secours.
push "dhcp-option DOMAIN m2l.local" # pousse le suffixe de recherche DNS. permet de taper uniquement "matchplanner" au lieu de "matchplanner.m2l.local".
```